SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)
Constituent of Symbiosis International (Deemed University), Pune
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॥वसुधैव कुटुम्बकम्॥ Re-Accredited by NAAC with 'A++' Grade

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Class : CSE
Batch : 2025-29

Assignment : 30

Subject: Programming with Java
ASSIGNMENT No : 30
TITLE : Write a JSP application to demonstrate CRUD operations.

Problem Statement : Write a JSP application to demonstrate CRUD operations.

Learning Objectives : To implement Create, Read, Update, and Delete operations using JSP and a database.

Theory :
CRUD stands for Create, Read, Update, and Delete, which represent the four basic database operations performed in web applications. JSP works together with Servlets and JDBC to implement these operations. Users interact with HTML forms created using JSP, while JDBC executes SQL statements to manipulate database records. INSERT adds new records, SELECT retrieves information, UPDATE modifies existing data, and DELETE removes unwanted records. Proper validation, exception handling, and session management improve application reliability and security. JSP CRUD applications are widely used in employee management systems, online shopping portals, hospital management systems, banking applications, and other enterprise software.

Code :

A30.jsp > html

```
<!DOCTYPE html>
<html>
<head>
    <title>JSP CRUD</title>
</head>
<body>

<h2>JSP CRUD Application</h2>

<p>CRUD stands for:</p>

<ul>
    <li>Create - Add records</li>
    <li>Read - View records</li>
    <li>Update - Modify records</li>
    <li>Delete - Remove records</li>
</ul>

<a href="Exp1.jsp">Employee Management</a>
<br><br>
<a href="Exp2.jsp">Patient Management</a>

</body>
</html>
```

SQL Pre-requisites :

Limit to 1000 rows

```
create database atharva1;
use atharva1;

CREATE TABLE employee (
    emp_id INT PRIMARY KEY,
    name VARCHAR(50),
    department VARCHAR(50),
    salary DOUBLE
);

CREATE TABLE patient (
    patient_id INT PRIMARY KEY,
    name VARCHAR(50),
    age INT,
    disease VARCHAR(100),
    treatment_status VARCHAR(50)
);
```

Output :



JSP CRUD Application

CRUD stands for:

- Create - Add records
- Read - View records
- Update - Modify records
- Delete - Remove records

Employee Management

Patient Management

Exercise :

1. **Create a JSP-based application to add, update, delete, and view employee details**

Ans) Code :

```
<%@ page import="java.sql.*" %>

<!DOCTYPE html>
<html>
<head>
   <title>Employee CRUD</title>
</head>
<body>

<h2>Employee Management</h2>

<form method="post">
   Employee ID:
   <input type="number" name="emp_id" required>
   <br><br>

   Name:
   <input type="text" name="name">
   <br><br>

   Department:
   <input type="text" name="department">
   <br><br>

   Salary:
   <input type="number" step="0.01" name="salary">
   <br><br>

   <input type="submit" name="action" value="Add">
   <input type="submit" name="action" value="Update">
   <input type="submit" name="action" value="Delete">
   <input type="submit" name="action" value="View">
</form>

<%
   String action = request.getParameter("action");
```

```
    if (action != null) {
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");

            Connection con = DriverManager.getConnection(
                "jdbc:mysql://localhost:3306/atharva1",
                "root",
                "Abcd@1234"
            );

            int id = Integer.parseInt(request.getParameter("emp_id"));

            if (action.equals("Add")) {
                PreparedStatement ps = con.prepareStatement(
                    "INSERT INTO employee (emp_id, name, department, salary) VALUES (?,
?, ?, ?)"
                );

                ps.setInt(1, id);
                ps.setString(2, request.getParameter("name"));
                ps.setString(3, request.getParameter("department"));
                ps.setDouble(4, Double.parseDouble(request.getParameter("salary")));

                ps.executeUpdate();
                out.println("<p>Employee added successfully.</p>");

                ps.close();

            } else if (action.equals("Update")) {
                PreparedStatement ps = con.prepareStatement(
                    "UPDATE employee SET name=?, department=?, salary=? WHERE
emp_id=?"
                );

                ps.setString(1, request.getParameter("name"));
                ps.setString(2, request.getParameter("department"));
                ps.setDouble(3, Double.parseDouble(request.getParameter("salary")));
                ps.setInt(4, id);

                int rows = ps.executeUpdate();
                out.println("<p>" + (rows > 0 ?
```

```
                "Employee updated successfully." :
                "Employee not found.") + "</p>");

            ps.close();

        } else if (action.equals("Delete")) {
            PreparedStatement ps = con.prepareStatement(
                "DELETE FROM employee WHERE emp_id=?"
            );

            ps.setInt(1, id);

            int rows = ps.executeUpdate();
            out.println("<p>" + (rows > 0 ?
                "Employee deleted successfully." :
                "Employee not found.") + "</p>");

            ps.close();

        } else if (action.equals("View")) {
            Statement stmt = con.createStatement();
            ResultSet rs = stmt.executeQuery(
                "SELECT * FROM employee"
            );

            out.println("<h3>Employee Records</h3>");
            out.println("<table border='1' cellpadding='6'>");
            out.println("<tr><th>ID</th><th>Name</th><th>Department</th><th>Salary
</th></tr>");

            while (rs.next()) {
                out.println("<tr>");
                out.println("<td>" + rs.getInt("emp_id") + "</td>");
                out.println("<td>" + rs.getString("name") + "</td>");
                out.println("<td>" + rs.getString("department") + "</td>");
                out.println("<td>" + rs.getDouble("salary") + "</td>");
                out.println("</tr>");
            }

            out.println("</table>");
```

```
            rs.close();
            stmt.close();
        }

        con.close();

    } catch (Exception e) {
        out.println("<p>Error: " + e.getMessage() + "</p>");
    }
}
%>

</body>
</html>
```

Output :

Employee Management

Employee ID: 101

Name: Vatsal

Department: CSE

Salary: 50000

Add Update Delete View

Employee Management

Employee ID:

Name:

Department:

Salary:

Add Update Delete View

Employee added successfully.

Employee Management

Employee ID: 101

Name: Vatsal C

Department: Entc

Salary: 51000

Add Update Delete View

Error: empty String

Employee Management

Employee ID: 105

Name: Fatima

Department: Entc

Salary:

Add Update Delete View

Employee Management

Employee ID:

Name:

Department:

Salary:

Add Update Delete View

Employee deleted successfully.

Employee Management

Employee ID:

Name:

Department:

Salary:

Add Update Delete View

Employee Records

ID	Name	Department	Salary
101	Vatsal C	Entc	51000.0
102	Atharva	Civil	80000.0
103	Rajesh	CSE	60000.0

2. Create a JSP-based application to manage patient records using CRUD operations. Include fields such as patient ID, name, age, disease, and treatment status.

Ans) Code :

```
<%@ page import="java.sql.*" %>

<!DOCTYPE html>
<html>
<head>
   <title>Patient CRUD</title>
</head>
<body>

<h2>Patient Management</h2>

<form method="post">
   Patient ID:
   <input type="number" name="patient_id" required>
   <br><br>

   Name:
   <input type="text" name="name">
   <br><br>

   Age:
   <input type="number" name="age">
   <br><br>

   Disease:
   <input type="text" name="disease">
   <br><br>

   Treatment Status:
   <input type="text" name="treatment_status">
   <br><br>

   <input type="submit" name="action" value="Add">
   <input type="submit" name="action" value="Update">
   <input type="submit" name="action" value="Delete">
```

```
    <input type="submit" name="action" value="View">
</form>

<%
    String action = request.getParameter("action");

    if (action != null) {
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");

            Connection con = DriverManager.getConnection(
                "jdbc:mysql://localhost:3306/atharva1",
                "root",
                "Abcd@1234"
            );

            int id = Integer.parseInt(
                request.getParameter("patient_id")
            );

            if (action.equals("Add")) {
                PreparedStatement ps = con.prepareStatement(
                    "INSERT INTO patient (patient_id, name, age, disease, treatment_status)
VALUES (?, ?, ?, ?, ?)"
                );

                ps.setInt(1, id);
                ps.setString(2, request.getParameter("name"));
                ps.setInt(3, Integer.parseInt(request.getParameter("age")));
                ps.setString(4, request.getParameter("disease"));
                ps.setString(5, request.getParameter("treatment_status"));

                ps.executeUpdate();
                out.println("<p>Patient added successfully.</p>");
                ps.close();

            } else if (action.equals("Update")) {
                PreparedStatement ps = con.prepareStatement(
                    "UPDATE patient SET name=?, age=?, disease=?, treatment_status=?
WHERE patient_id=?"
                );
```

```
            ps.setString(1, request.getParameter("name"));
            ps.setInt(2, Integer.parseInt(request.getParameter("age")));
            ps.setString(3, request.getParameter("disease"));
            ps.setString(4, request.getParameter("treatment_status"));
            ps.setInt(5, id);

            int rows = ps.executeUpdate();
            out.println("<p>" + (rows > 0 ?
               "Patient updated successfully." :
               "Patient not found.") + "</p>");
            ps.close();

         } else if (action.equals("Delete")) {
            PreparedStatement ps = con.prepareStatement(
               "DELETE FROM patient WHERE patient_id=?"
            );

            ps.setInt(1, id);

            int rows = ps.executeUpdate();
            out.println("<p>" + (rows > 0 ?
               "Patient deleted successfully." :
               "Patient not found.") + "</p>");
            ps.close();

         } else if (action.equals("View")) {
            Statement stmt = con.createStatement();
            ResultSet rs = stmt.executeQuery(
               "SELECT * FROM patient"
            );

            out.println("<h3>Patient Records</h3>");
            out.println("<table border='1' cellpadding='6'>");
            out.println("<tr><th>ID</th><th>Name</th><th>Age</th><th>Disease</th><
th>Treatment Status</th></tr>");

            while (rs.next()) {
               out.println("<tr>");
               out.println("<td>" + rs.getInt("patient_id") + "</td>");
               out.println("<td>" + rs.getString("name") + "</td>");
```

```
                out.println("<td>" + rs.getInt("age") + "</td>");
                out.println("<td>" + rs.getString("disease") + "</td>");
                out.println("<td>" + rs.getString("treatment_status") + "</td>");
                out.println("</tr>");
            }

            out.println("</table>");
            rs.close();
            stmt.close();
        }

        con.close();

    } catch (Exception e) {
        out.println("<p>Error: " + e.getMessage() + "</p>");
    }
  }
%>

</body>
</html>
```

Output :

Patient Management

Patient ID: 101

Name: Raj

Age: 21

Disease: Diabetes

Treatment Status: Diagnosed

Add Update Delete View

Patient Management

Patient ID:

Name:

Age:

Disease:

Treatment Status:

Add Update Delete View

Patient added successfully.

Patient Management

Patient ID: 101

Name: Raj

Age: 21

Disease: Diabetes

Treatment Status: Cured

Add Update Delete View

Patient added successfully.

Patient Management

Patient ID:

Name:

Age:

Disease:

Treatment Status:

Add Update Delete View

Patient updated successfully.

Patient Management

Patient ID: 103

Name: John

Age:

Disease:

Treatment Status:

Add Update Delete View

Patient Management

Patient ID:

Name:

Age:

Disease:

Treatment Status:

Add Update Delete View

Patient deleted successfully.

Patient Management

Patient ID:

Name:

Age:

Disease:

Treatment Status:

Add Update Delete View

Patient Records

ID	Name	Age	Disease	Treatment Status
101	Raj	21	Diabetes	Cured
102	Arhan	22	Dengue	Under Observation
104	Ragini	28	T.B	On Meds
105	Zaid	44	Cancer	Chemotherapy

Conclusion : Thus, the JSP applications successfully demonstrate CRUD (Create, Read, Update, and Delete) operations on employee and patient records using JDBC and MySQL. They allow users to add, view, update, and delete records stored in the database, providing an efficient way to manage employee details and patient information through a web interface.